Graph Entry and Closure of the Selected-Minimizer Route

This note implements the remaining Plan 1 regularity step after the single-set selection principle. It turns the compactness phrase "for the selected minimizer close enough to the ball" into an explicit deterministic pipeline:

$$\begin{aligned}\alpha(U) \text{ small} &\Longrightarrow d_H(\partial U, \partial B_1) \text{ small}\\ &\Longrightarrow \text{Alt–Caffarelli flatness at a fixed scale}\\ &\Longrightarrow \partial U \text{ is a global graph} \Longrightarrow E(U) - E(B_1) \gtrsim \alpha(U).\end{aligned}$$

The constants are not numerical, but every threshold is named and depends only on the fixed data (N, R) and the BDV free-boundary constants.

1 Selected-Minimizer Input

Let $U = \{u > 0\} \subset B_R$ be a selected minimizer from `quantitative-selection-principle.md`, with $\tau \leq \tau_{\text{reg}}(N, R)$. After translating by the barycenter, assume the comparison ball is B_1.

The BDV regularity package gives constants

$$c_d, r_d, L_u, q_-, q_+, L_q > 0$$

depending only on (N, R), such that:

1. two-sided density holds at every free-boundary point and every $r \leq r_d$;
2. u is Lipschitz with constant L_u;
3. u is nondegenerate:
$$\sup_{B_r(x)} u \geq c_d r \qquad (x \in \partial U,\ r \leq r_d);$$
4. once the free boundary is flat, the Bernoulli coefficient satisfies
$$q_- \leq q_u \leq q_+, \qquad \operatorname{Lip} q_u \leq L_q.$$

These are exactly the constants entering BDV Lemmas 4.9, 4.12, 4.16 and Theorem 4.18.

2 From Asymmetry to Hausdorff Distance

Let

$$M := |U \Delta B_1|.$$

The density estimate implies

$$d_H(\partial U, \partial B_1) \leq C_H(N, R, c_d, r_d) M^{1/N}.$$

Since BDV Lemma 4.2 gives

$$|U \Delta B_1| \leq C_\alpha(N) \alpha(U)^{1/2},$$

we get

$$\boxed{d_H(\partial U, \partial B_1) \leq C'_H(N,R)\alpha(U)^{1/(2N)}.}$$

The proof is purely geometric: if a free-boundary point lies distance d from ∂B_1, a ball of radius comparable to d lies on one side of ∂B_1, while density forces a chunk of $U \Delta B_1$ of size $\gtrsim d^N$. The reverse inclusion is identical, using the density of the two phases.

3 Hausdorff Closeness Gives Flatness

Let $\bar{\mu}, \bar{\rho}, \gamma, C_{\mathrm{AC}}$ be the constants in the Alt–Caffarelli improvement theorem used by BDV, with the above bounds on u and q_u. Fix

$$0 < \mu \leq \frac{\bar{\mu}}{16 C_0},$$

where C_0 is the harmless constant converting the geometric slab below into the $F(\cdot, 1, +\infty)$ normalization. Choose

$$0 < \rho_\mu \leq \bar{\rho}$$

so small that every cap of ∂B_1 of radius $6\rho_\mu$ lies in a $\mu\rho_\mu$-slab around its tangent plane:

$$\partial B_1 \cap B_{6\rho_\mu}(\bar{x}) \subset \{x : |(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq \mu\rho_\mu\}.$$

Define

$$h_F := \frac{\mu\rho_\mu}{16}.$$

Proposition 3.1: fixed-scale flatness. If

$$d_H(\partial U, \partial B_1) \leq h_F,$$

then for every $\bar{x} \in \partial B_1$ there is $x_0 \in \partial U \cap B_{h_F}(\bar{x})$ such that, in $B_{4\rho_\mu}(x_0)$, the function u is of Alt–Caffarelli class

$$F(C_0\mu, 1, +\infty)$$

with respect to the tangent direction $\nu_{\bar{x}}$.

Proof. The Hausdorff inclusion gives

$$\partial U \cap B_{5\rho_\mu}(\bar{x}) \subset \{x : |(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq \mu\rho_\mu + h_F\}.$$

Moving the center from $\bar{x}$ to x_0 changes the slab height by at most h_F, hence

$$\partial U \cap B_{4\rho_\mu}(x_0) \subset \{|(x - x_0) \cdot \nu_{\bar{x}}| \leq 2\mu\rho_\mu\}.$$

The density and nondegeneracy estimates give the two phase alternatives in the slabs outside this strip: points a fixed multiple of $\mu\rho_\mu$ inside the ball phase are positive with the lower linear growth required in the F-definition, while points the same distance in the exterior phase vanish. The Lipschitz bound for u supplies the upper linear control. These are precisely the three inequalities in BDV Definition 4.17, after multiplying μ by the universal conversion constant C_0.

4 Local Flatness Gives a Global Graph

By the choice of μ, $C_0\mu \le \bar{\mu}$. BDV Theorem 4.18 therefore improves the free boundary in every ball $B_{\rho_\mu}(x_0)$ above to a $C^{1,\gamma}$ graph with norm at most

$$C_{\mathrm{AC}}\mu.$$

Choose μ smaller, if needed, so that

$$C_{\mathrm{AC}}\mu \le \mu_{\mathrm{tub}}(N),$$

where $\mu_{\mathrm{tub}}(N)$ is a fixed tubular-neighborhood slope threshold for radial projection onto ∂B_1. Then the local graphs cannot fold over the radial direction. Since the balls $B_{\rho_\mu}(\bar{x})$ cover ∂B_1, the local parametrizations patch to one global function $g : \partial B_1 \to \mathbb{R}$:

$$\partial U = \{(1 + g(\theta))\theta : \theta \in \partial B_1\}.$$

Moreover

$$\|g\|_{L^\infty(\partial B_1)} \le C\, d_H(\partial U, \partial B_1),$$

and

$$\|g\|_{C^{1,\gamma}(\partial B_1)} \le C(N,R)\mu.$$

Higher $C^{k,\gamma}$ bounds follow from the explicit smooth selected Bernoulli law, with BDV Lemma 4.16 supplying the starting uniform coefficient bounds, and Schauder bootstrapping. These bounds are uniform, not small. Small C^{2,γ_0} norm is obtained by interpolating the uniform higher bound with the small L^∞ bound above.

5 Explicit Graph-Entry Threshold

With the constants above fixed, set

$$\alpha_{\mathrm{graph}}(N,R) := \left(\frac{h_F}{C'_H(N,R)}\right)^{2N}.$$

Then

$$\boxed{\alpha(U) \le \alpha_{\mathrm{graph}}(N,R) \quad \Longrightarrow \quad \partial U \text{ is a global } C^{1,\gamma} \text{ graph over } \partial B_1.}$$

This is the quantitative replacement for the compactness step in BDV.

6 Entry into the Nearly Spherical Theorem

BDV's nearly spherical theorem requires small C^{2,γ_0}, not merely small $C^{1,\gamma}$. Fix $0 < \gamma_0 < \gamma$ and an integer $m \ge 3$. The Schauder bootstrap gives

$$\|g\|_{C^{m,\gamma}(\partial B_1)} \le M_m(N,R).$$

Interpolation on ∂B_1 gives

$$\|g\|_{C^{2,\gamma_0}} \le C\|g\|_{L^\infty}^{1-\theta} M_m(N,R)^\theta.$$

Since $\|g\|_{L^\infty} \le C d_H(\partial U, \partial B_1) \le C\alpha(U)^{1/(2N)}$, there is a second threshold $\alpha_{\mathrm{sph}}(N,R,\gamma_0)$ such that

$$\alpha(U) \le \alpha_{\mathrm{sph}} \quad \Longrightarrow \quad \|g\|_{C^{2,\gamma_0}} \le \delta_{\mathrm{sph}}(N,\gamma_0).$$

Set

$$\alpha_{\mathrm{small}} := \min\{\alpha_{\mathrm{graph}}, \alpha_{\mathrm{sph}}\}.$$

7 Closure by the Nearly Spherical Theorem

Once

$$\partial U = \{(1 + g(\theta))\theta : \theta \in \partial B_1\},$$

the BDV nearly spherical expansion applies after volume and barycenter normalization. There is $c_{\mathrm{sph}}(N) > 0$ such that, for $\alpha(U) \le \alpha_{\mathrm{small}}$,

$$E(U) - E(B_1) \ge c_{\mathrm{sph}}(N)\|g\|^2_{H^{1/2}(\partial B_1)}.$$

The BDV asymmetry satisfies

$$\alpha(U) \le C_{\mathrm{sph}}(N)\|g\|^2_{L^2(\partial B_1)} \le C_{\mathrm{sph}}(N)\|g\|^2_{H^{1/2}(\partial B_1)}.$$

Therefore

$$\boxed{E(U) - E(B_1) \ge c_*(N)\alpha(U)}$$

for every selected minimizer with $\tau \le \tau_{\mathrm{reg}}$ and $\alpha(U) \le \alpha_{\mathrm{small}}$.

8 Final Selected-Counterexample Contradiction

Let $\Omega_0 \subset B_R$, $|\Omega_0| = |B_1|$, be an alleged bad set with

$$q := Q_\alpha(\Omega_0) = \frac{E(\Omega_0) - E(B_1)}{\alpha(\Omega_0)}$$

small. Apply the single-set selection map with $\tau = q^{1/4}$ and obtain the volume-normalized selected set $\widetilde{\Omega}$. The selection principle gives

$$Q_\alpha(\widetilde{\Omega}) \le 2C_{\mathrm{sel}}q,$$

and

$$\frac{1}{2}\alpha(\Omega_0) \le \alpha(\widetilde{\Omega}) \le \frac{3}{2}\alpha(\Omega_0).$$

If q and $\alpha(\Omega_0)$ are small enough that $\tau \le \tau_{\mathrm{reg}}$ and $\alpha(\Omega_0) \le 2\alpha_{\mathrm{small}}/3$, then $\alpha(\widetilde{\Omega}) \le \alpha_{\mathrm{small}}$, so the graph-entry theorem, the interpolation entry, and the nearly spherical estimate give

$$Q_\alpha(\widetilde{\Omega}) \ge c_*(N).$$

Thus no bad set can have

$$q < \frac{c_*(N)}{2C_{\mathrm{sel}}}.$$

Equivalently, in the small-asymmetry regime,

$$E(\Omega) - E(B_1) \ge c\,\alpha(\Omega).$$

Since BDV Lemma 4.2 gives $\mathcal{A}(\Omega)^2 \lesssim \alpha(\Omega)$, this is the sharp quadratic Fraenkel stability estimate.